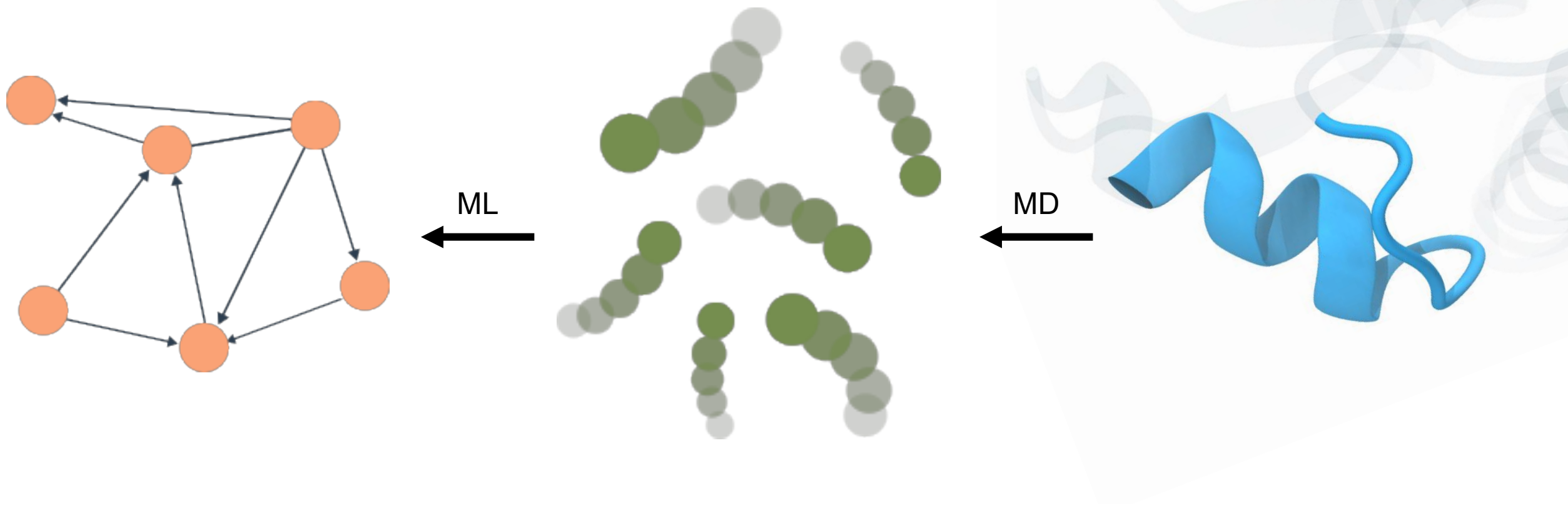
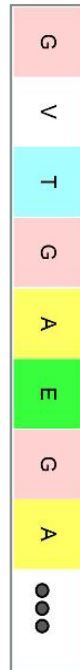


Learning **relational interactions** from **observed dynamics** for **biological systems**



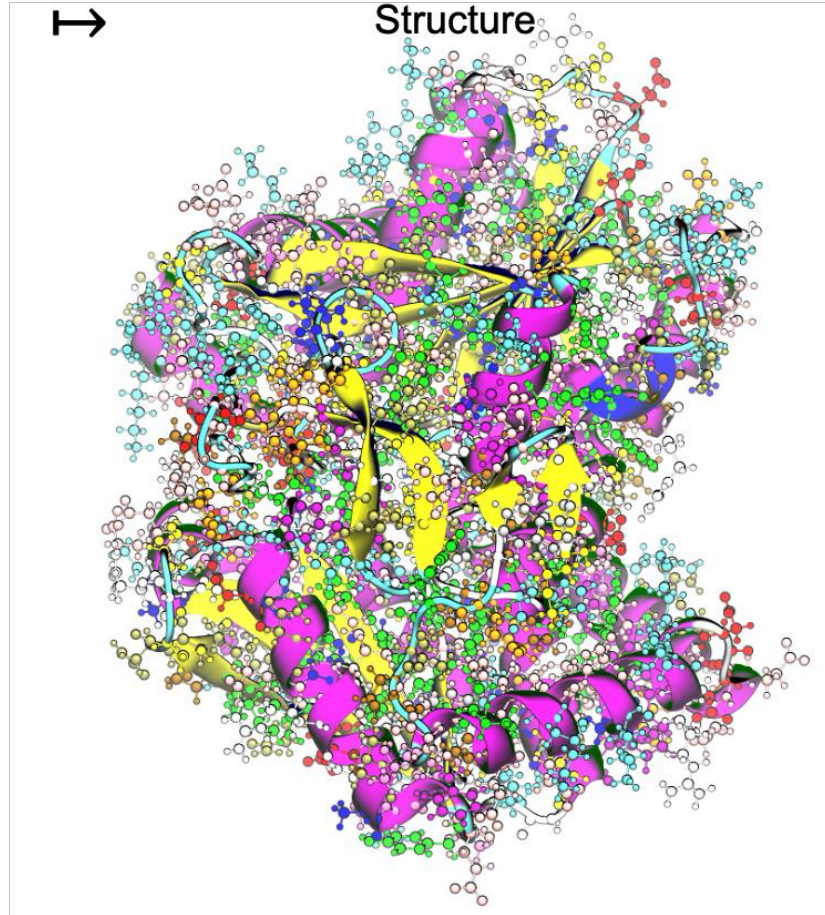
Motivation: Sequence, Structure, Function relationship in proteins

Sequence



→

Structure



→

Function

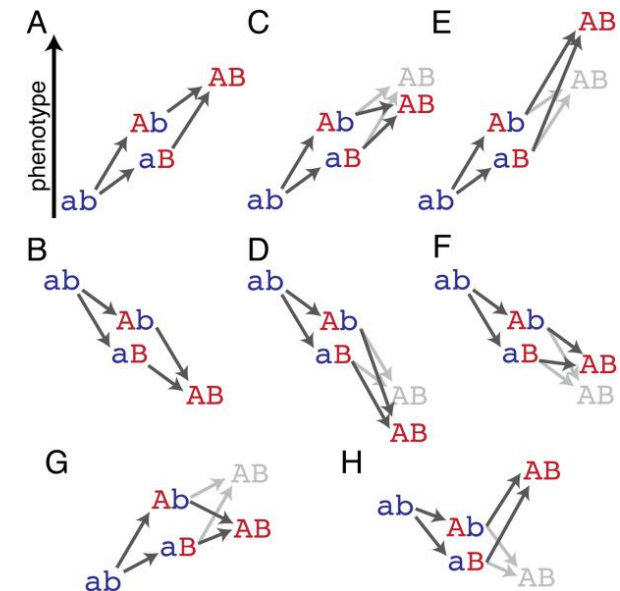
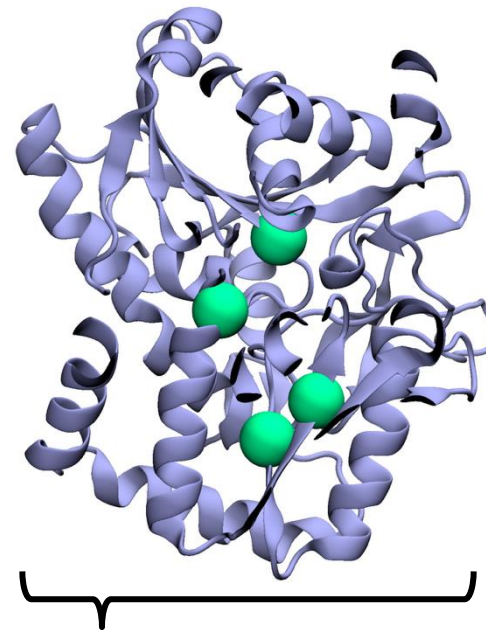
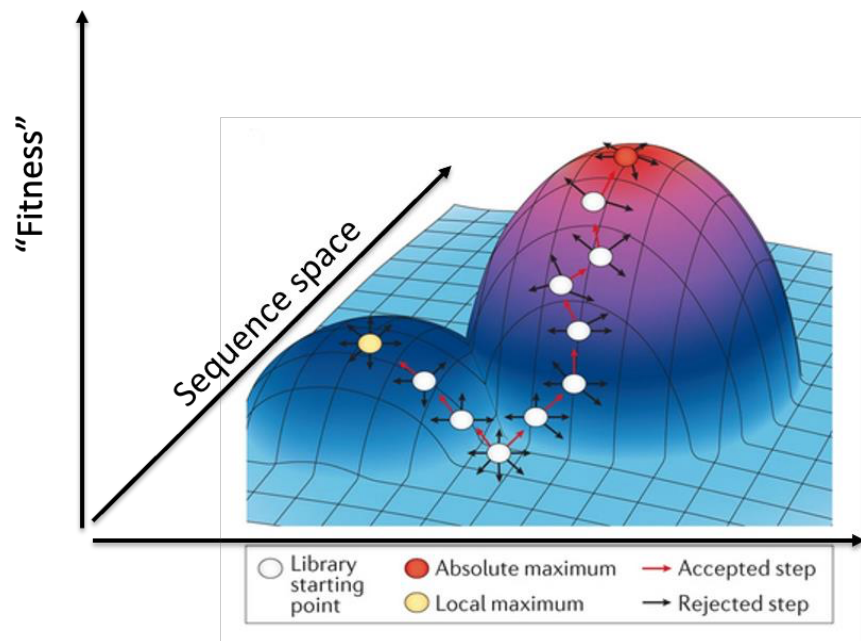
T_{50} ,

k_{cat} ,

K_D ,

...

Motivation: engineering approaches for enzyme design using natural proteins as starting points

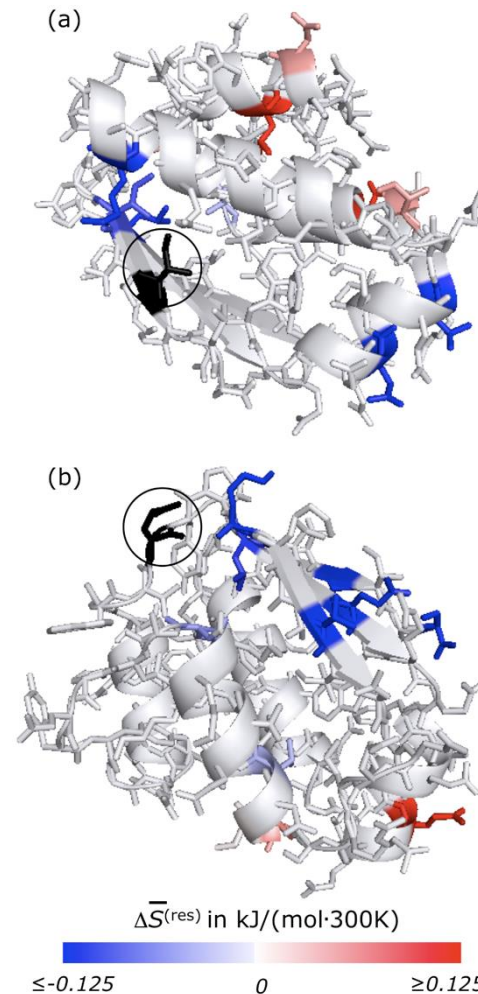
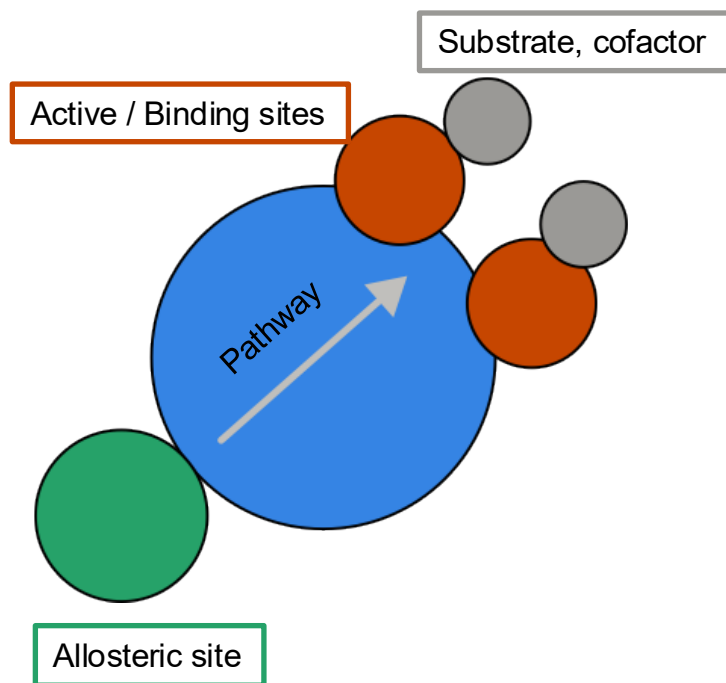


4 selected sites, 20 possible monomers $\rightarrow 20^4$ variants to test

Strategies for library design:

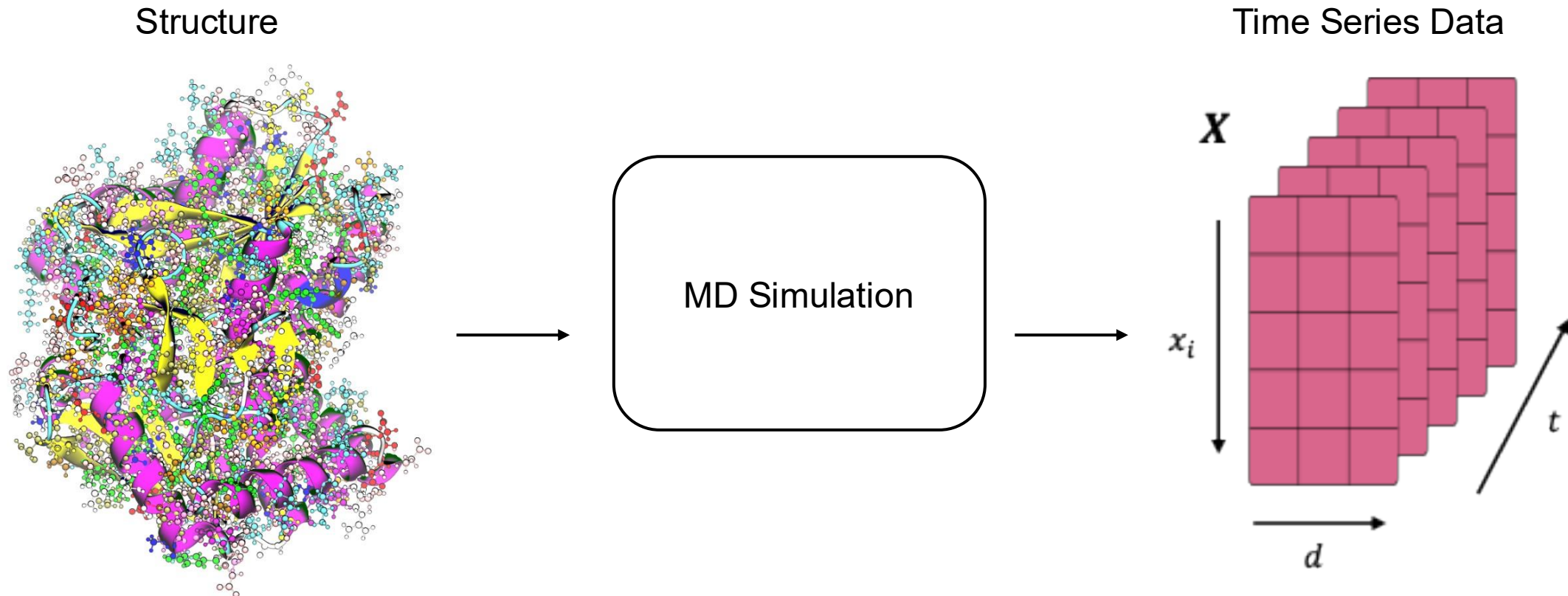
- Random selection
- Structural interactions
- Conserved pair-wise evolutionary data

Motivation: characterizing protein dynamics for actionable insights

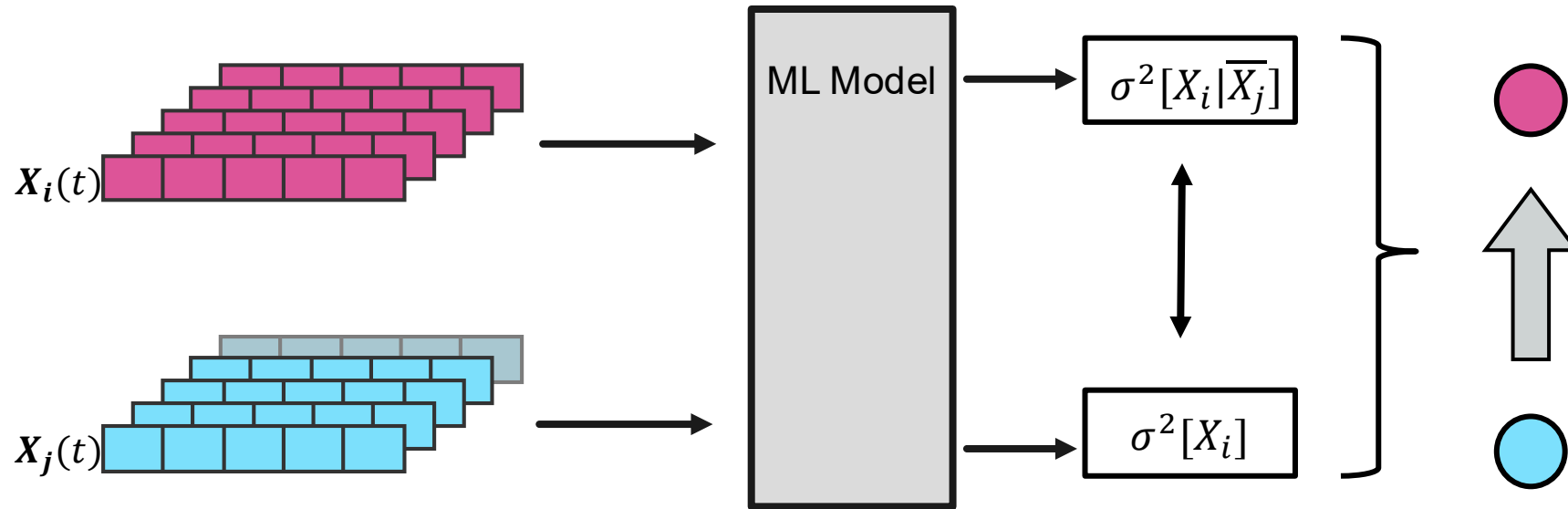


(Influence of mutation on explored dynamics / statistical ensemble of side chains)

What data do we observe from molecular dynamics simulations?



ML models for learning relational interactions



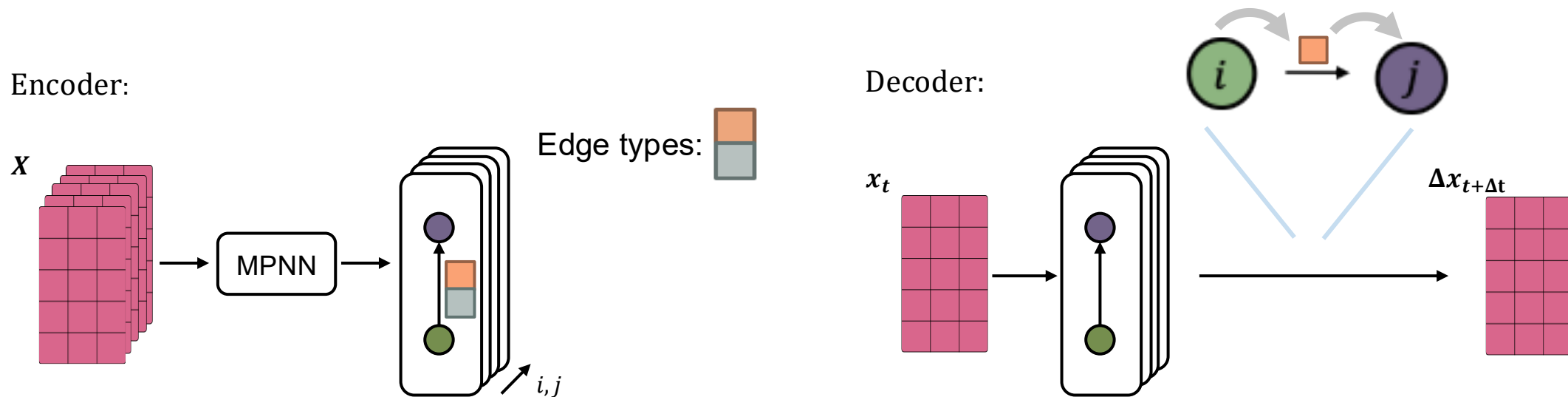
Granger, Clive WJ. "Investigating causal relations by econometric models and cross-spectral methods." *Econometrica: journal of the Econometric Society* (1969): 424-438.

Kipf, T.; Fetaya, E.; Wang, K.-C.; Welling, M.; Zemel, R. Neural Relational Inference for Interacting Systems. arXiv June 6, 2018.

<https://doi.org/10.48550/arXiv.1802.04687>.

Inferring interactions via a graph neural network ansatz

- Encoder: inputs observed timeseries and proposes a directed interaction graph to summarize dynamics
- Decoder: tests usefulness of proposed interactions by predicting time-propagated dynamics

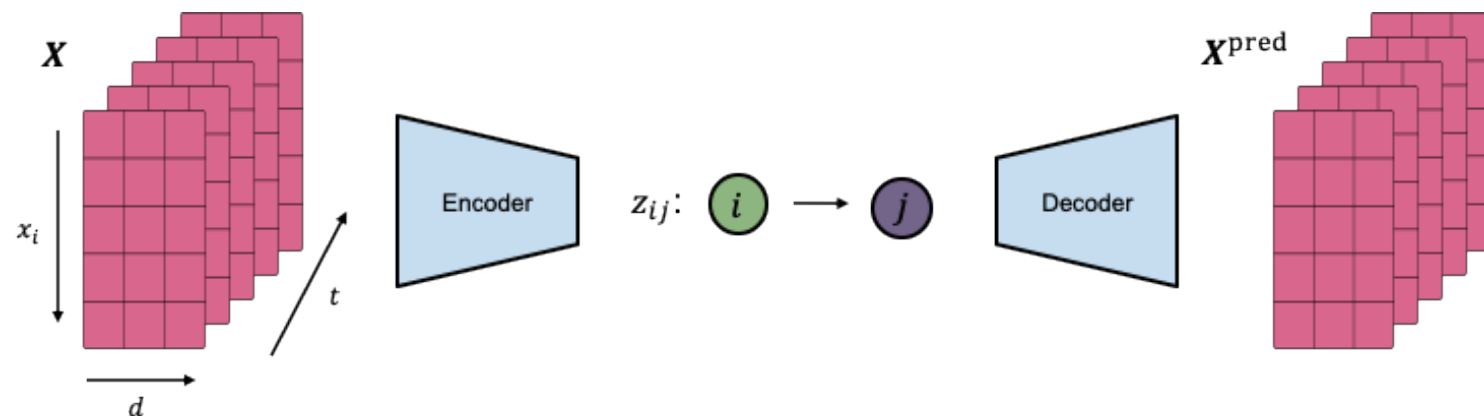


Workflow of learning relational interactions through simulation and model training

Generate Data:

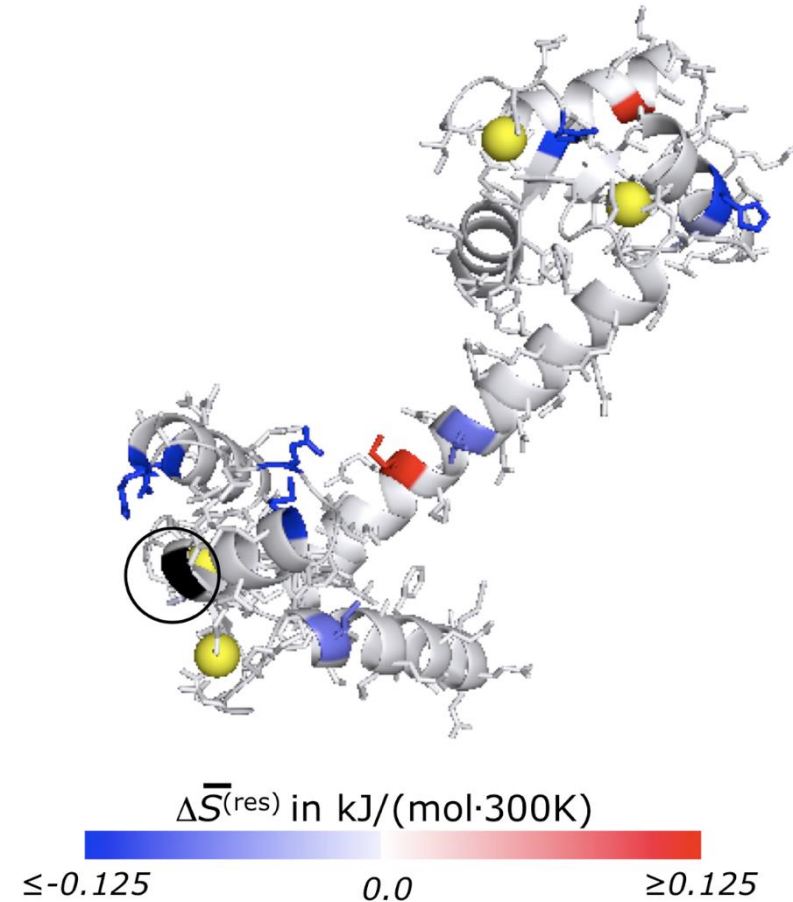


Train Model:

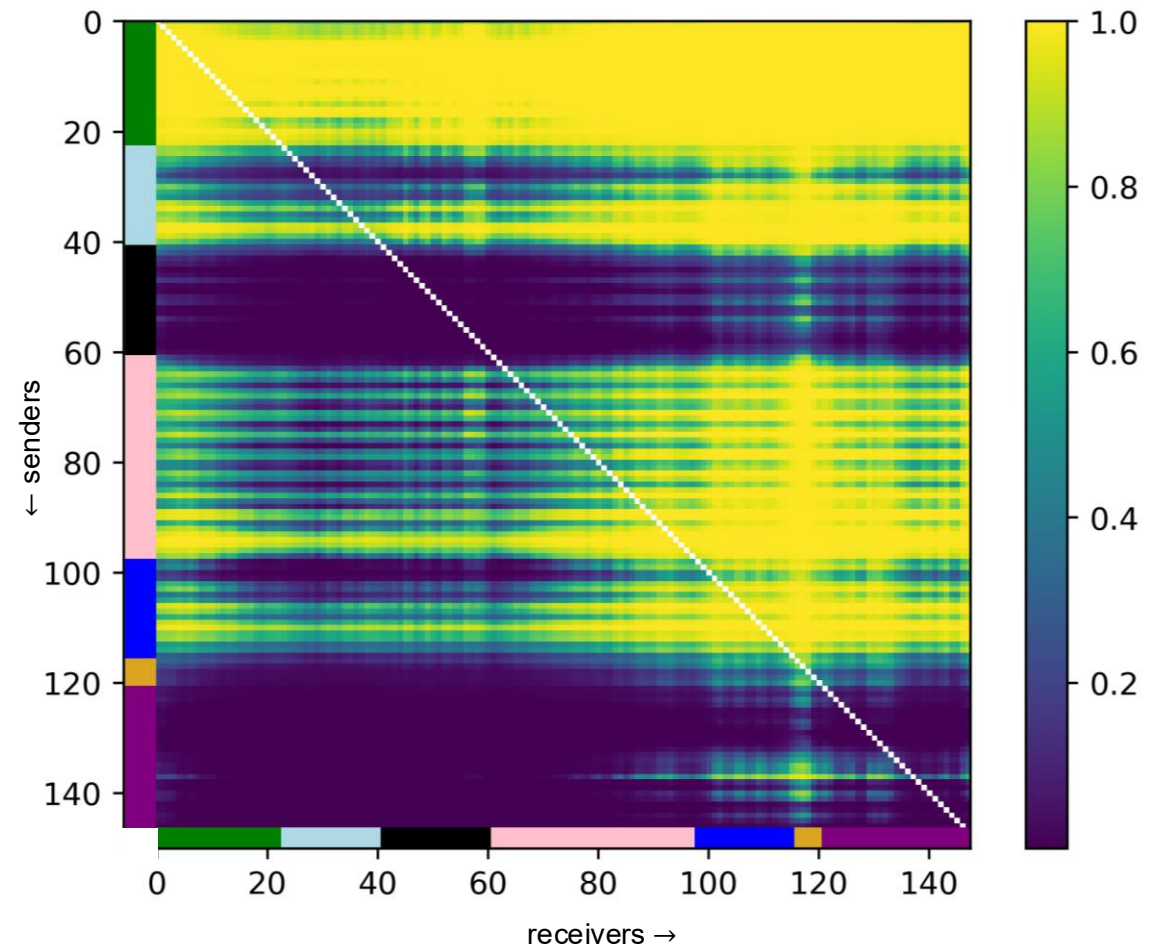
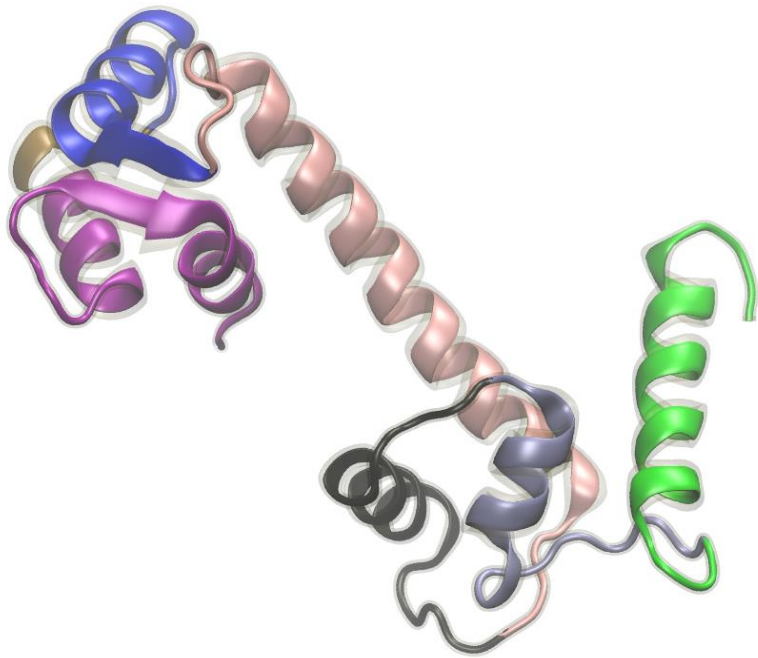


Test system: CaM protein system

- Calmodulin (CaM): highly conserved system in eukaryotic cells
- Small(er) structure (148 residues)
- Calcium detecting protein
 - Undergoes allosterically regulated conformational change upon calcium binding
 - Positive cooperativity between the 4 binding sites (complex interacting behavior at long range)



Trained GNN predictive edge results for CaM system

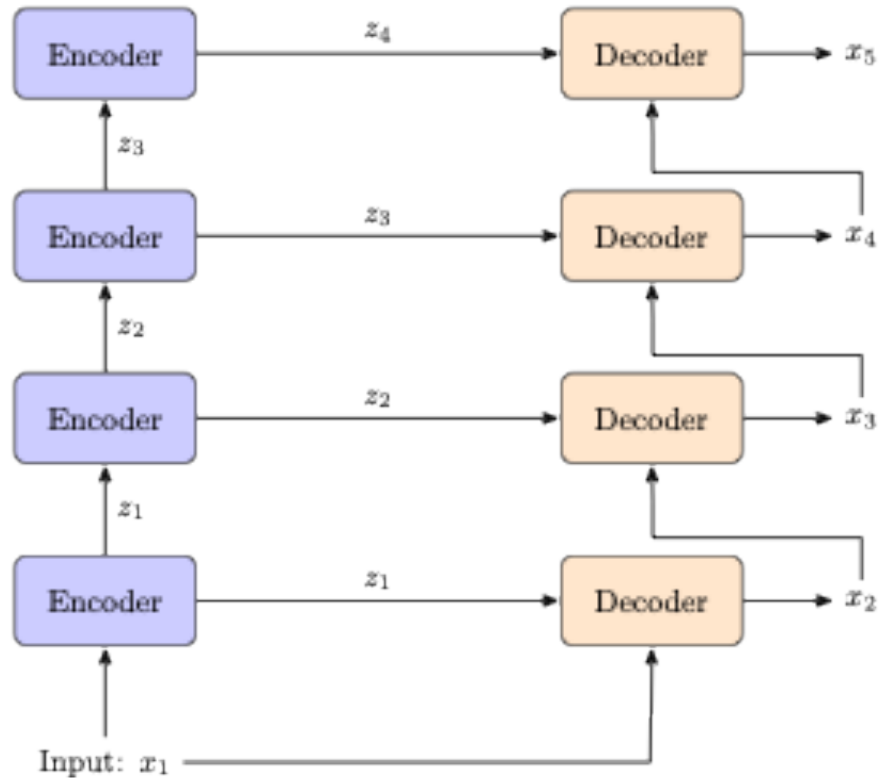


Shortcomings of GNN approach

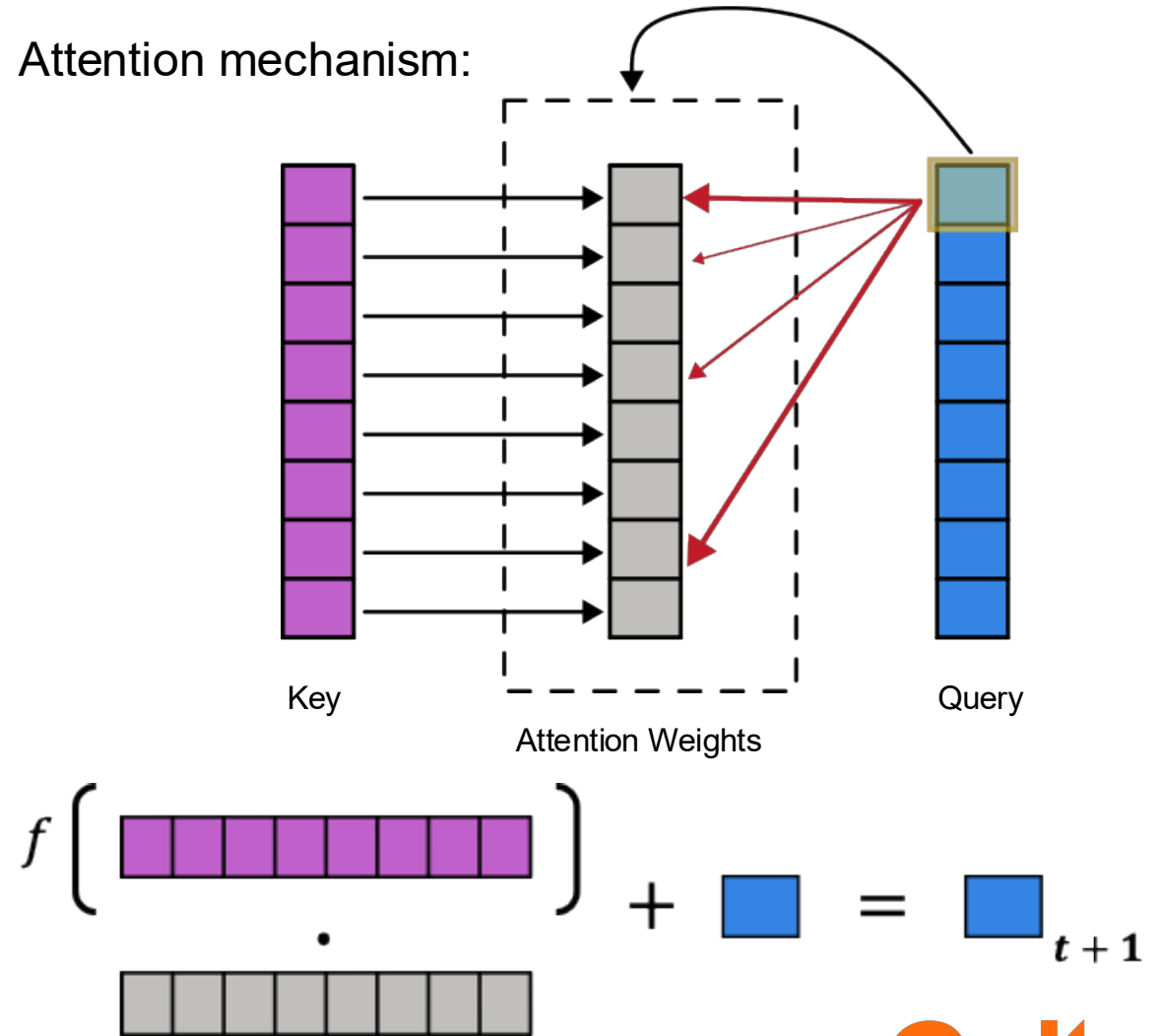
- GNN model does not incorporate particle identities
- The encoder model lacks expressivity
- The model scales poorly with system size and training dataset size
 - For 150 residue system and 1000 training examples with batch size of 8 on 32 GB V100 NVIDIA GPU → 2 minutes per epoch
 - For 380 residue system → 50 minutes per epoch

An alternative approach using attention mechanisms

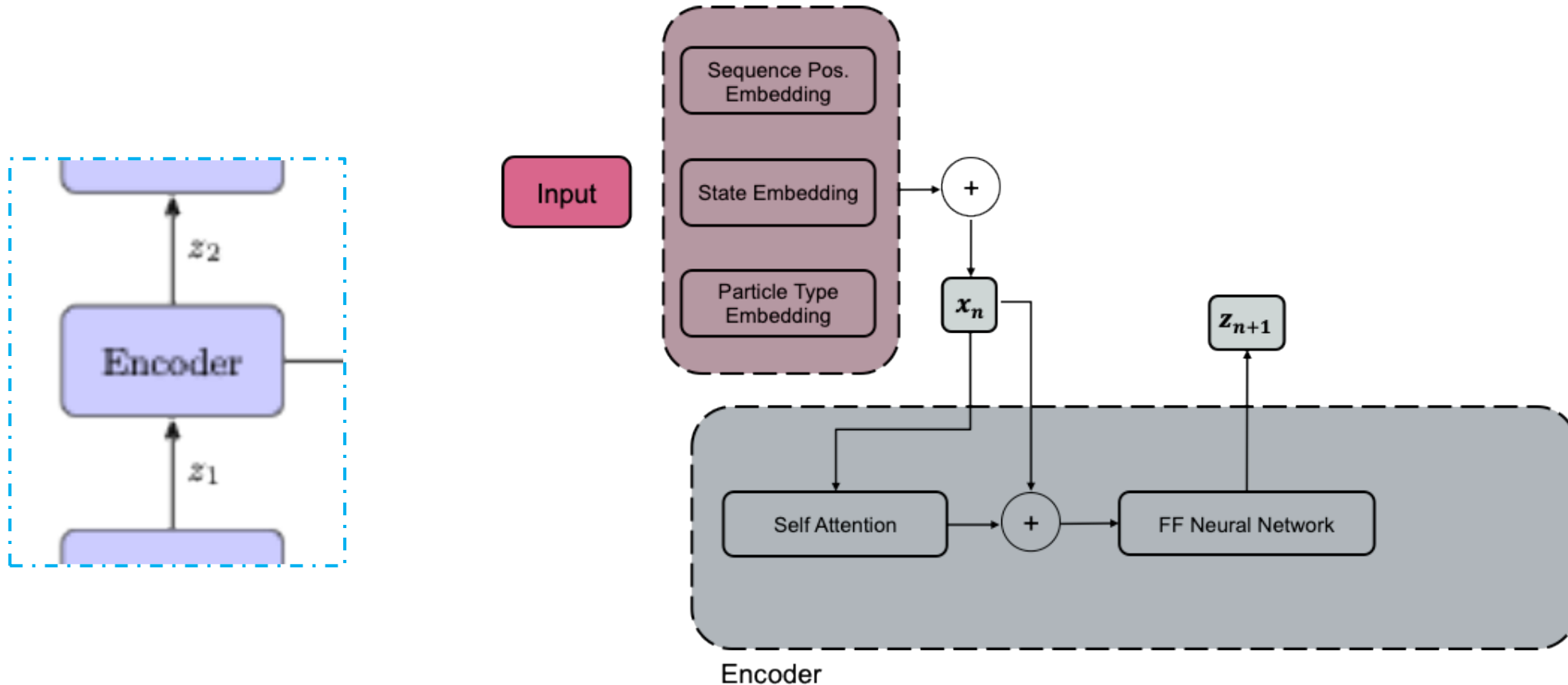
Stacked Transformer Model:



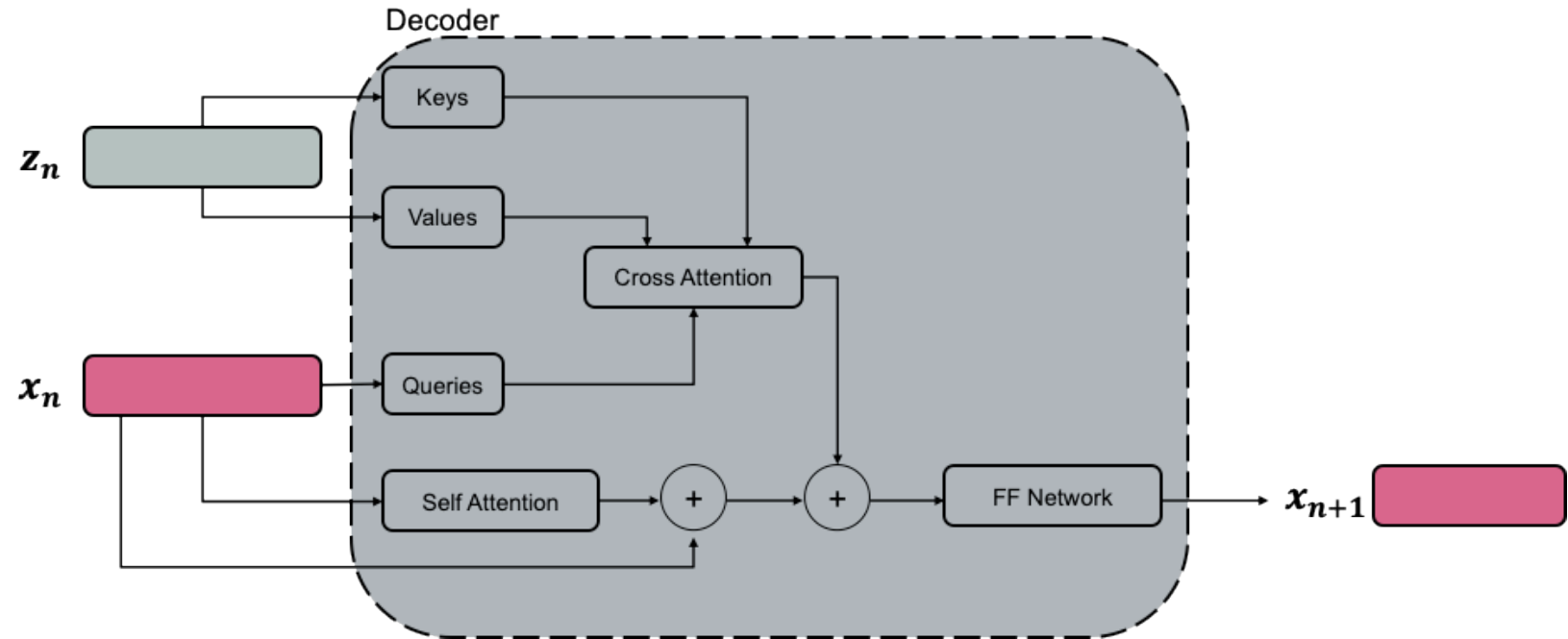
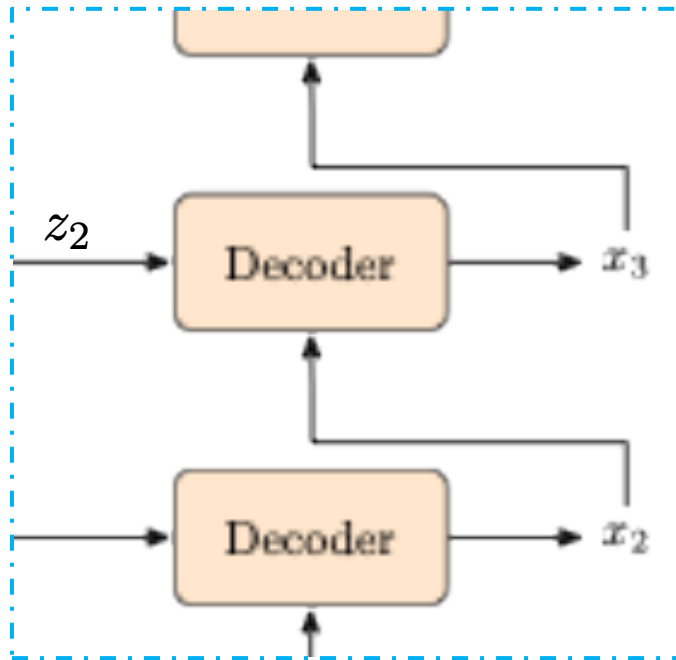
Attention mechanism:



Stacked transformer architecture: encoder



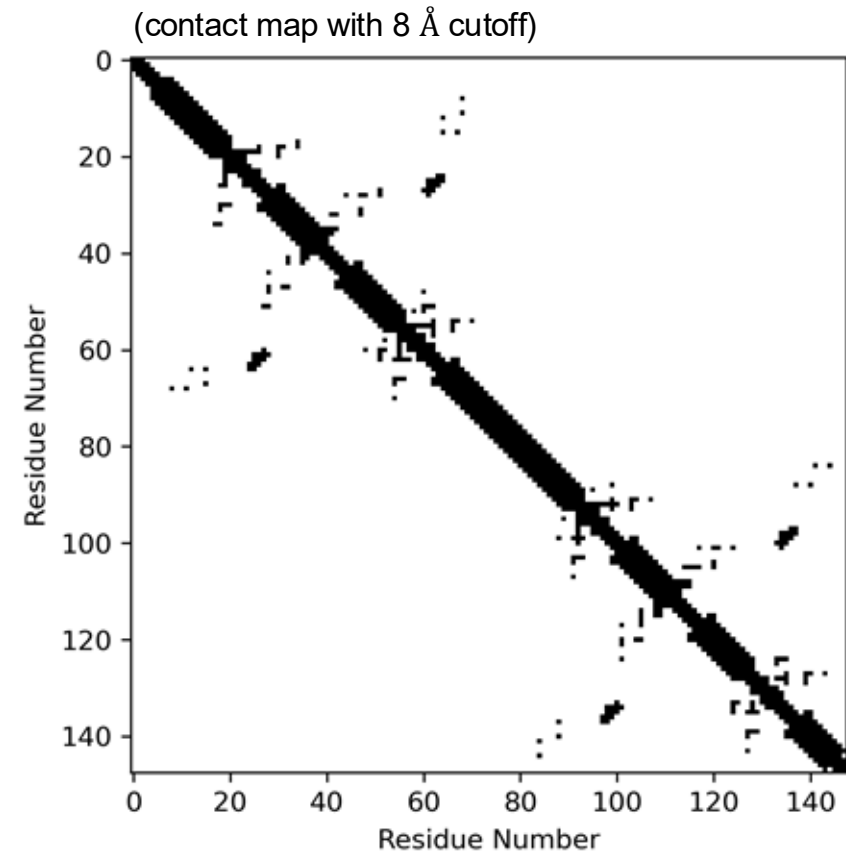
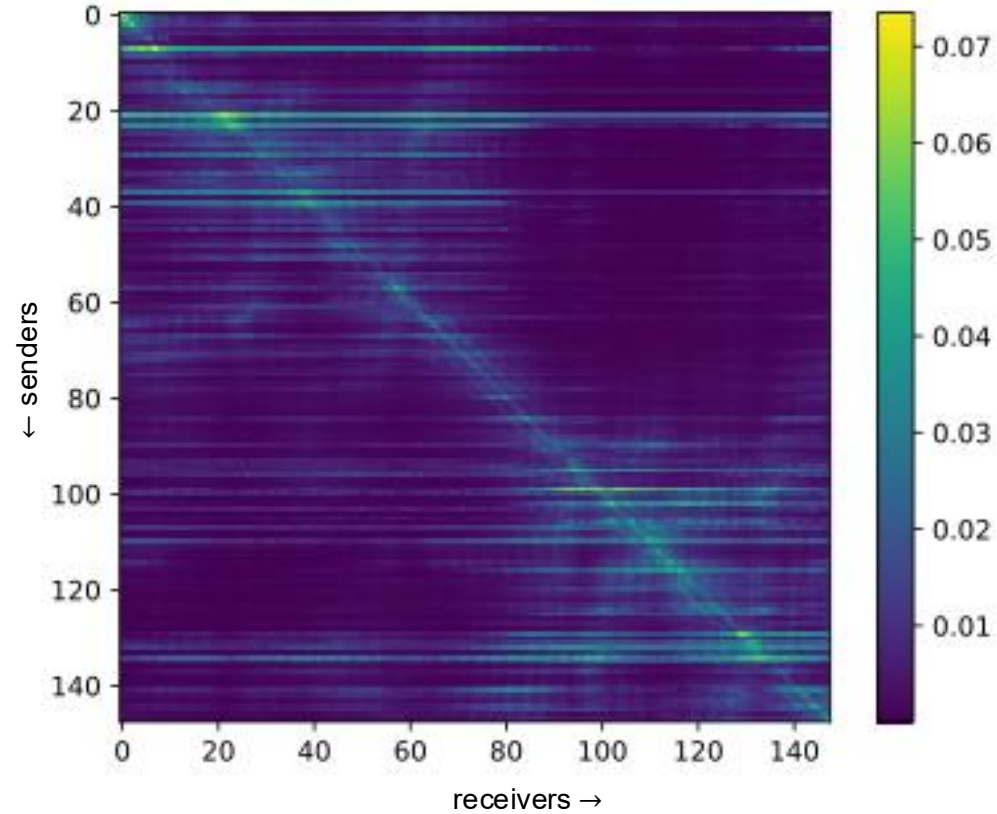
Stacked transformer architecture: decoder



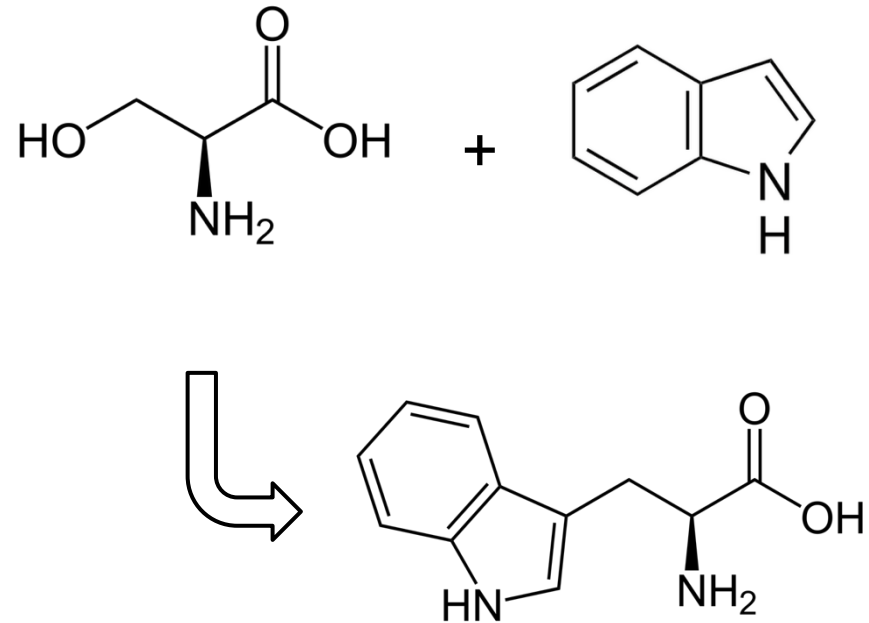
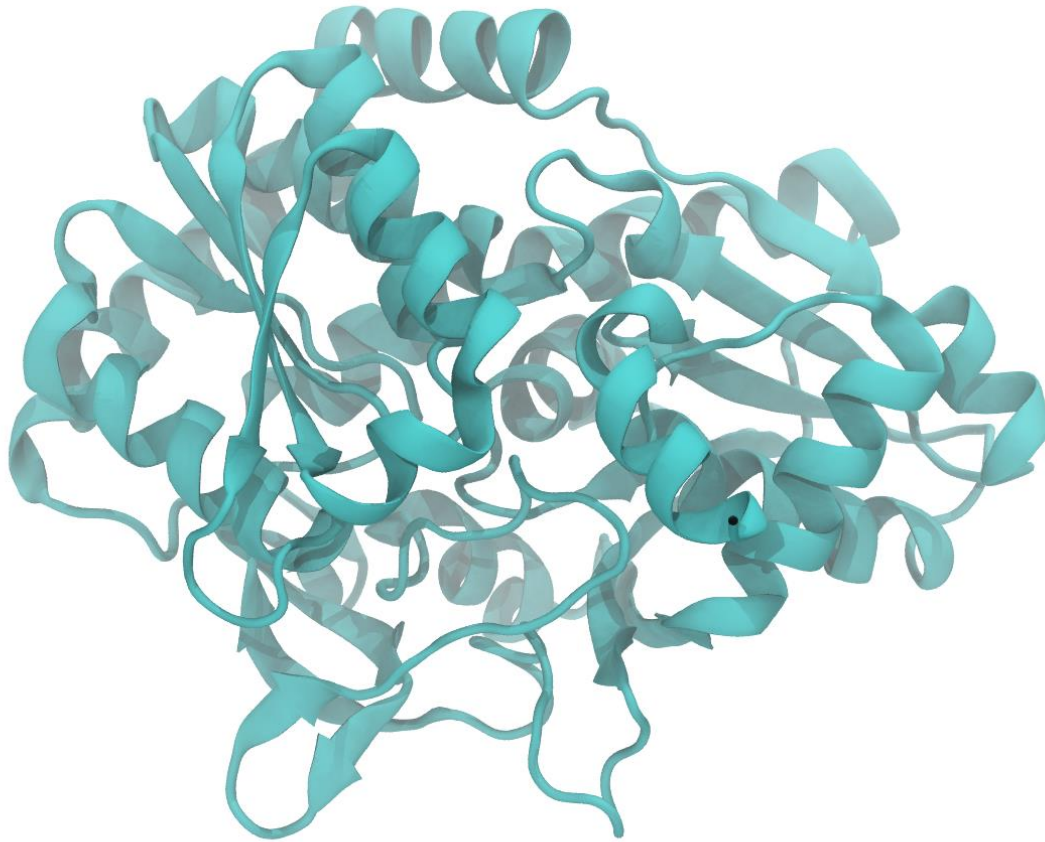
Improvements accessed by transformer model

- Scaling:
 - CaM system (150 residues) → 10x speedup
 - 380 residue system → 100x speedup (feasible to train)
- Increased expressivity
- Can include in embedding sequence position and residue identity

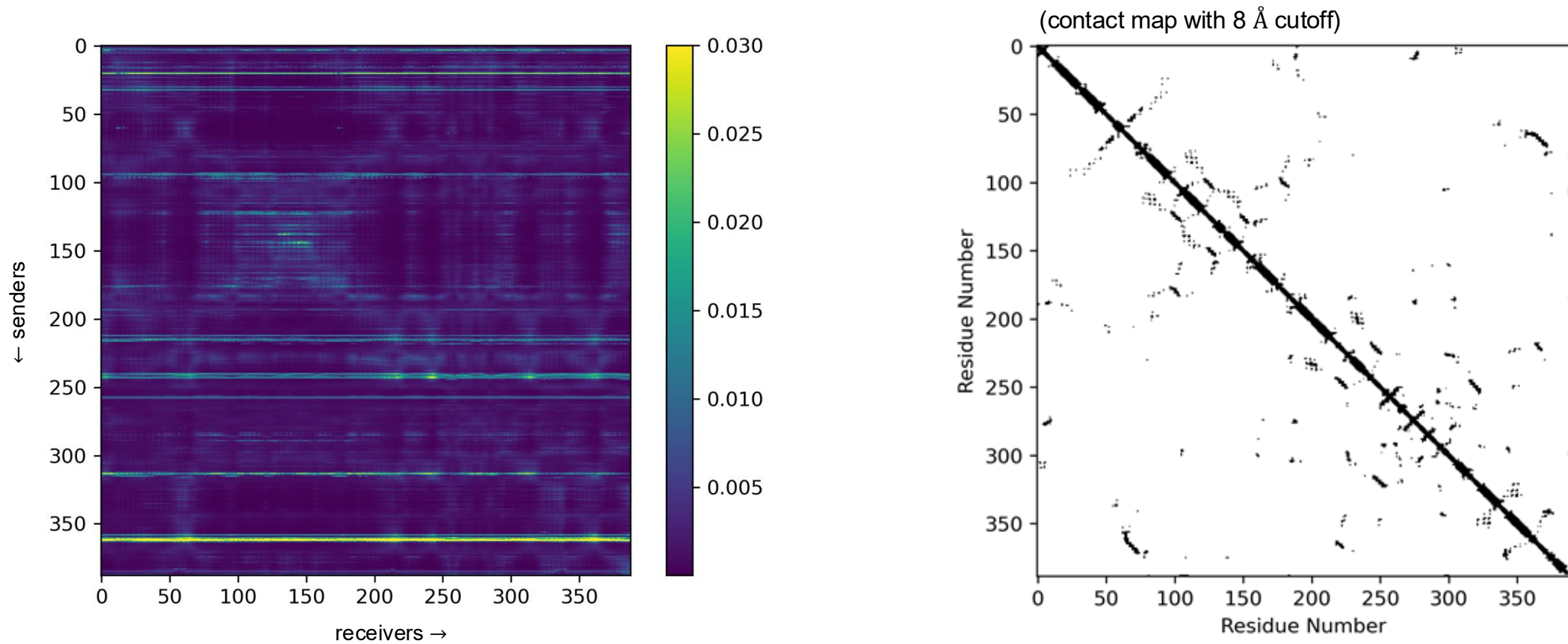
Transformer predictions for CaM dynamics



Moving to a larger system of interest: β -subunit of the Tryptophan Synthase enzyme

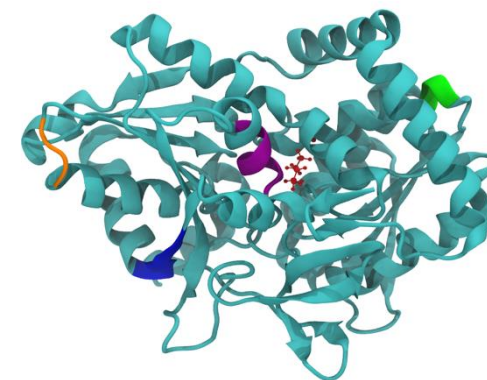
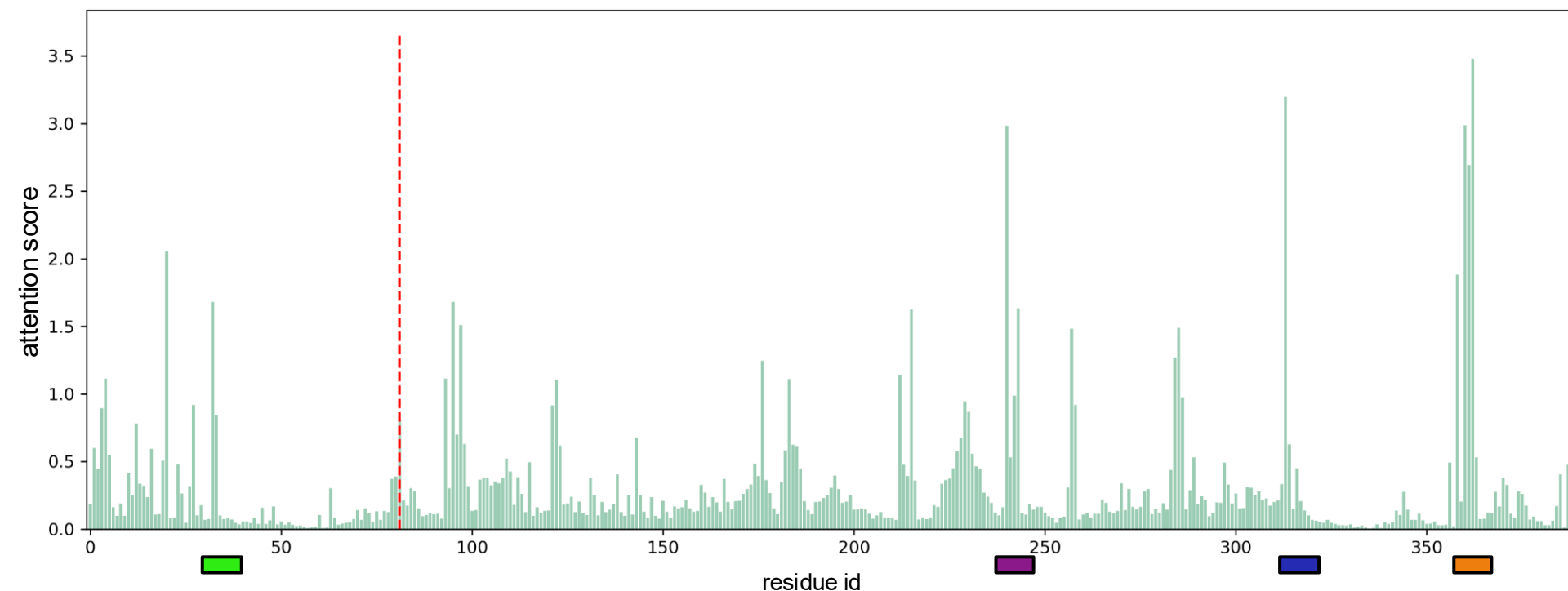


Transformer predictions for TrpB dynamics



Viewing the attention map for the catalytic lysine

Attention patterns for catalytic lysine residue:



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